

$$\begin{array}{lcl}
 a & = & \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
 b & = & \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\
 j & = & \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}
 \end{array}$$